

Clustering of meteorological data to improve agricultural decisions: a case study with SIMAGRO-RS

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ABSTRACT ORIGINAL

Context: Climate variability plays a fundamental role in shaping agricultural outcomes, influencing crop growth, yield, and resource management strategies. Problem: Identifying meaningful patterns in meteorological datasets to optimize agricultural practices is a challenging and complex task. Solution: This paper explores the use of clustering techniques on SIMAGRO-RS weather data to uncover meaningful patterns for agricultural applications. IS Theory: This study addresses the challenges of Information Systems applied to Agriculture, especially related to General Systems Theory (TGS). Method: The data was, first, extracted, then, preprocessed, filtered, aggregated, and standardized. Clustering was performed using K-means, K-medoids, and Hierarchical Agglomerative Clustering algorithms to cluster data from 19 meteorological stations in groups of stations with similar climate behavior. To define the number of clusters, the Elbow Plot and Silhouette Score methods were used. After clustering, the clusters were evaluated to identify the best result in terms of intracluster similarity. Supervised learning was used to design a specific metric for this problem, using an RMSE error measure. Summary of Results: From the experiments, the number of clusters $k = 2$ was found as the optimal number for the dataset. Regarding the clustering algorithm, the K-medoids algorithm presented better performance, with lower prediction RMSE (0.4990). IS Contributions and Impact: Based on the results, ways of interpreting clustering are suggested, in order to make it useful in the planning of agricultural practices in the state of Rio Grande do Sul. © 2024 ACM.